Lab 6 Push Notification

Key steps followed to make it work

Step 1: Created the flutter Application
New flutter project was created named fmc_notification_app using the following terminal command.

Step 2: Set up firebase project
- Navigate to console.firebase.google.com and sign in
- Clicked "Add project" and named it FMC Notification App
- Register an Android app
- Downloaded google_service.json and placed it inside android/app

Step 3. Configured Gradle files
Two Gradles file needed to be modified for firebase to work
- Added Google Service class path
- Added plugin and updated minSdk

Step 4. Added flutter depencies

Step 5. Configured AndroidManifest.xml.
Added requires permission to android/app/src/main/androidManifest.XML

Step 6. Initialized firebase in main.dart

Step 7. Requested Notification Permission.
the app requests permission from user on first launch.

Step 8: Retrieved and displayed Device token
The FCM device token was retrieved and displayed in the app UI

Step 9: Implemented Notification Handling
Three notification scenarios were implemented
- Foreground (app is open)
- Background (app minized)
- Terminated state (app closed)

Step 10: Sent Test notification from firebase console

① Navigated to firebase console → Devops & Eng → Messaging
② Clicked 'Create your first campaign' → Firebase Notification M
③ Enter notification title and body text
④ Clicked 'Send test message' and paste the token device
⑤ Clicked test.
⑥ Successfully received the notification on real Android device

My observation

① Foreground Notification Behaviour
When the application was open in foreground, firebase did not automatically display the notification as a system banner on android.

② Background Notification Behaviour
When the application was minimized to the background, Firebase handled the notification entirely on its own and displayed it correctly in the android notification bar ~~triggered~~ without any additional code

③ Permission request on Androind 13+
On Android 13 (API level 33) and above, the app must explicitly request the POST_NOTIFICATION permission at runtime using firebaseMessage.requestPermission(). Without this step, all notification are silently blocked by android system.

④ Gradle configuration is critical
The firebase integration depends entirely on the correct Gradle configuration. The google-service.json file must be placed in android/app/ specifically, not android/ alone

Screenshots — Application

1 App Launch — Device Token Displayed

The app launches and immediately fetches the FCM device token, displaying it on screen inside the Device Token card. A Copy Token button allows easy copying for use in Firebase Console.

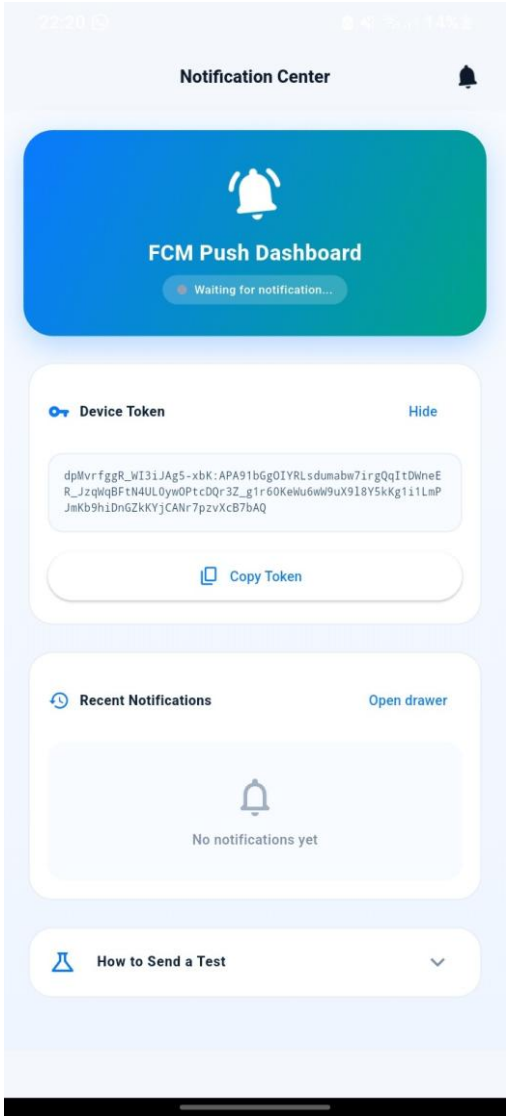

Figure 1: FCM Push Dashboard showing the device token on first launch

2 Foreground Notification with AlertDialog Popup

When a notification is received while the app is open, an AlertDialog popup appears immediately with the notification title and body. The status banner at the top shows 'Received in Foreground' and a local notification banner also appears at the top of the screen.

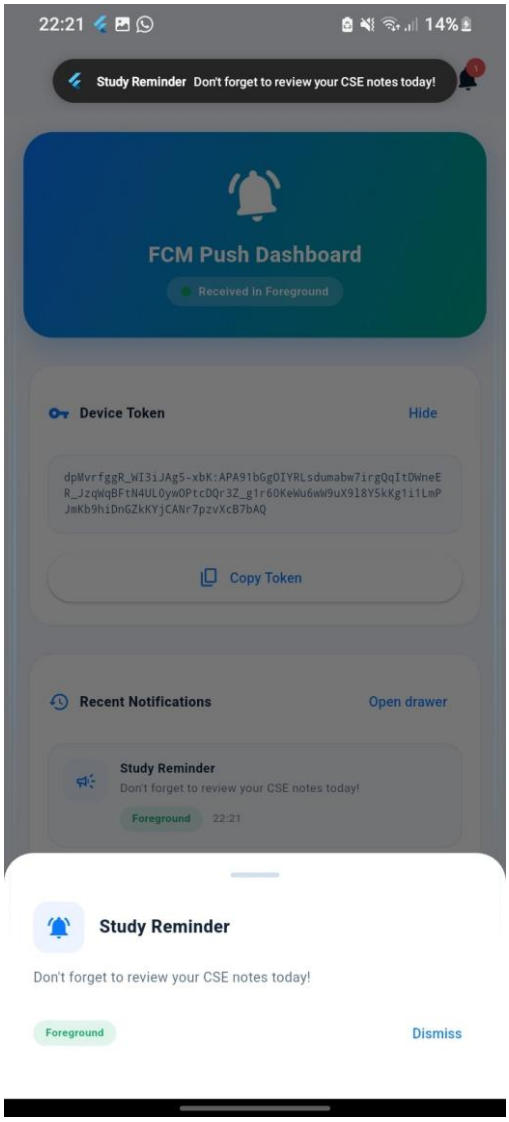

Figure 2: AlertDialog popup shown when a foreground notification is received

3 Recent Notifications List in App UI

The app maintains a list of all received notifications shown in the Recent Notifications section. Each entry shows the notification title, body, source (Foreground/Background), and timestamp. The notification badge count is visible on the app icon in the status bar.

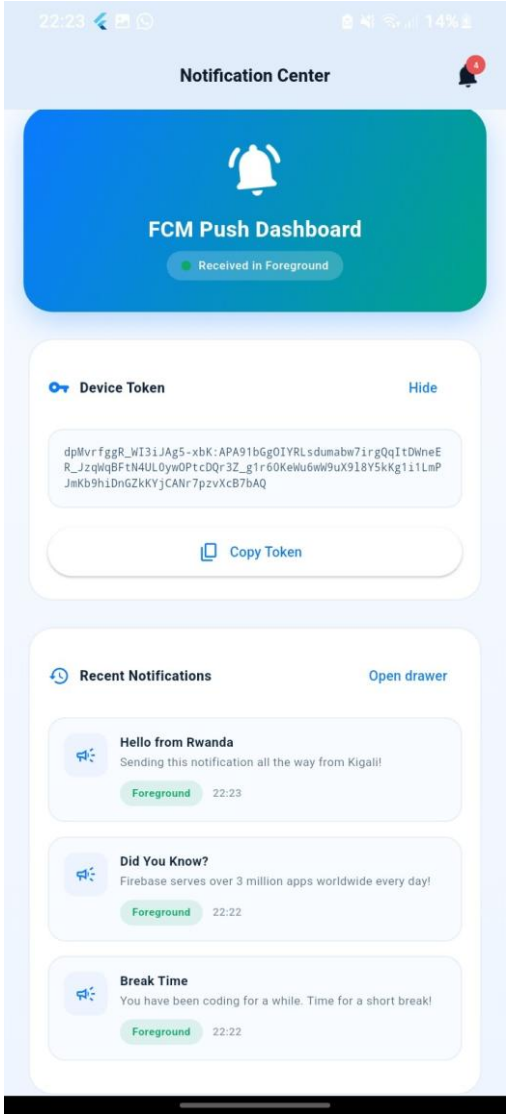

Figure 3: App UI showing recent notifications received in foreground

4 System Notification Bar — Background State

When the app is in the background, Firebase automatically displays the notifications in the Android system notification bar. Multiple notifications from the FCM Notification App are grouped together and visible to the user.

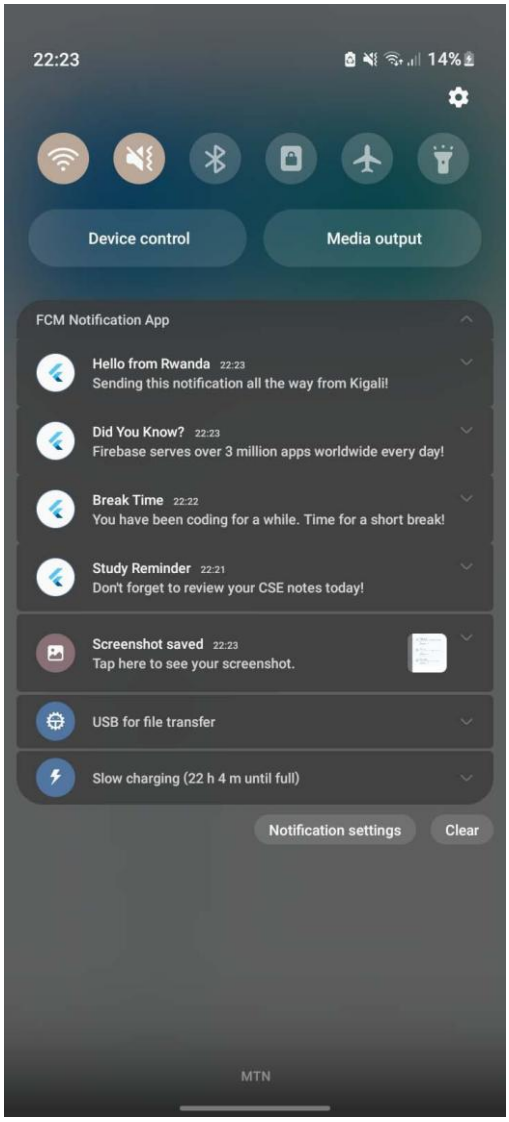

Figure 4: Android notification bar showing multiple received push notifications

5 In-App Notification Drawer

The app includes a Notification Drawer that shows all 4 notifications received during testing. Each card displays the notification title, body, source tag, and time of receipt.

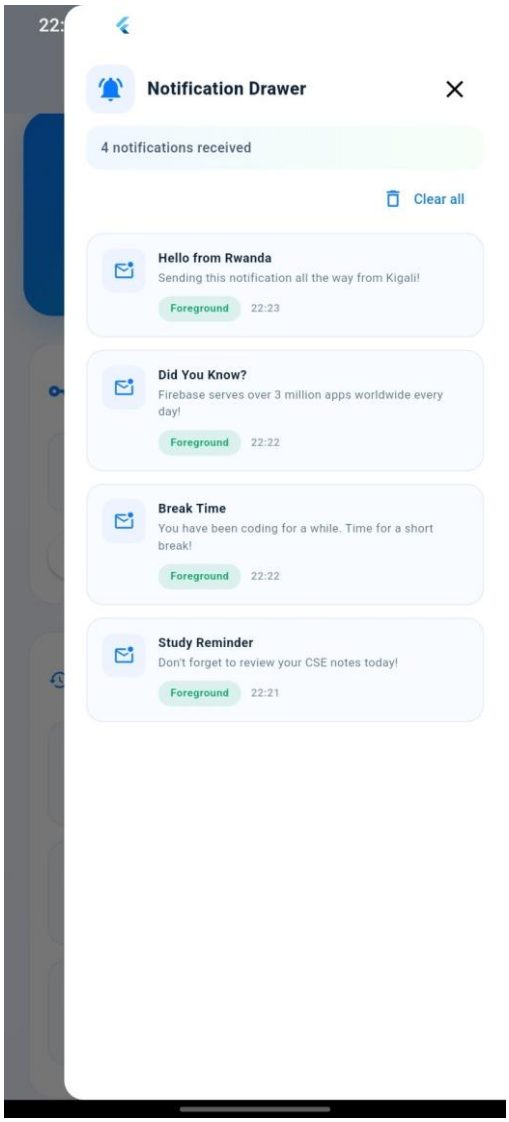

Figure 5: In-app notification drawer showing all received notifications

Screenshots — Firebase Cloud Messaging Console

1 Firebase Project Overview

The Firebase project named FCM Notification App was created under the Spark (free) plan. The project shows 2 registered apps and the Cloud Messaging service is active with a previously sent test notification visible in the Run section.

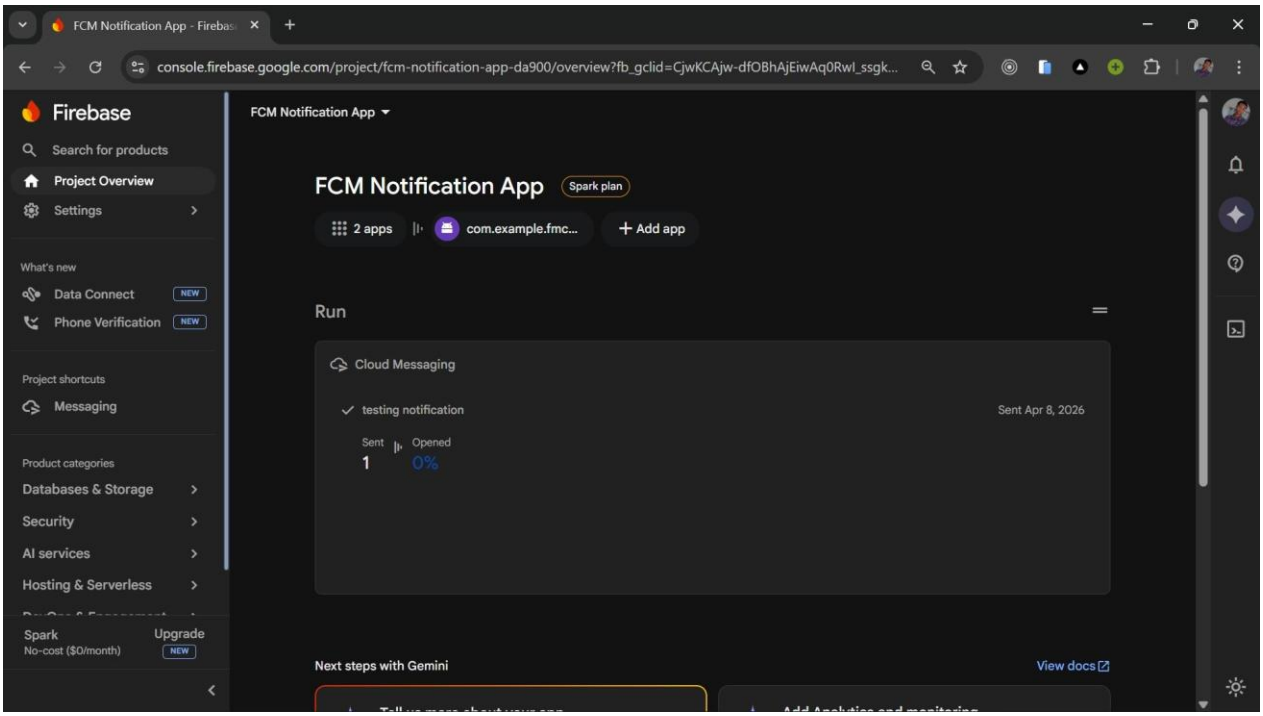

Figure 6: Firebase project overview showing the FCM Notification App project

2 Firebase Messaging — Campaigns Dashboard

The Firebase Cloud Messaging Campaigns tab shows the successfully completed test notification campaign named 'testing notification', which was sent on April 8, 2026. The status shows Completed with 1 send recorded.

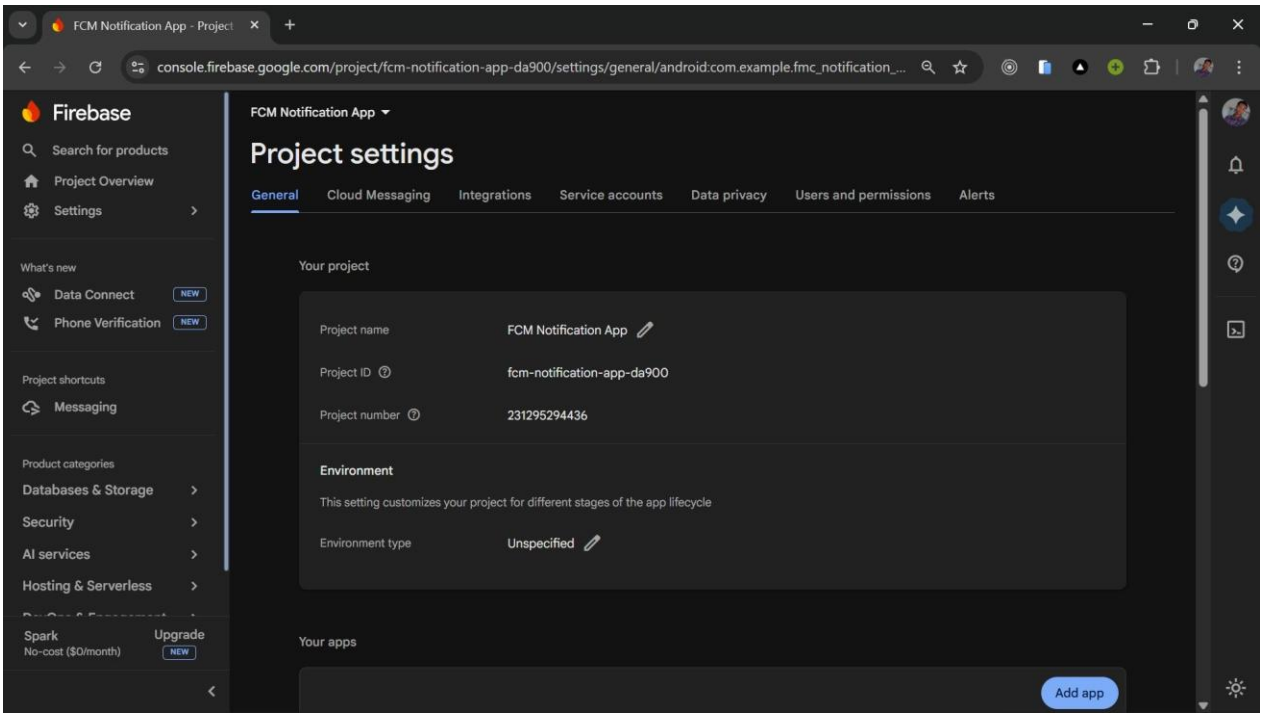

Figure 7: Firebase Messaging dashboard showing the completed notification campaign

3 Compose Notification Interface

The Compose Notification screen in Firebase Console where the notification title, body text, and optional image URL are entered. A live device preview on the right shows how the notification will appear on the device.

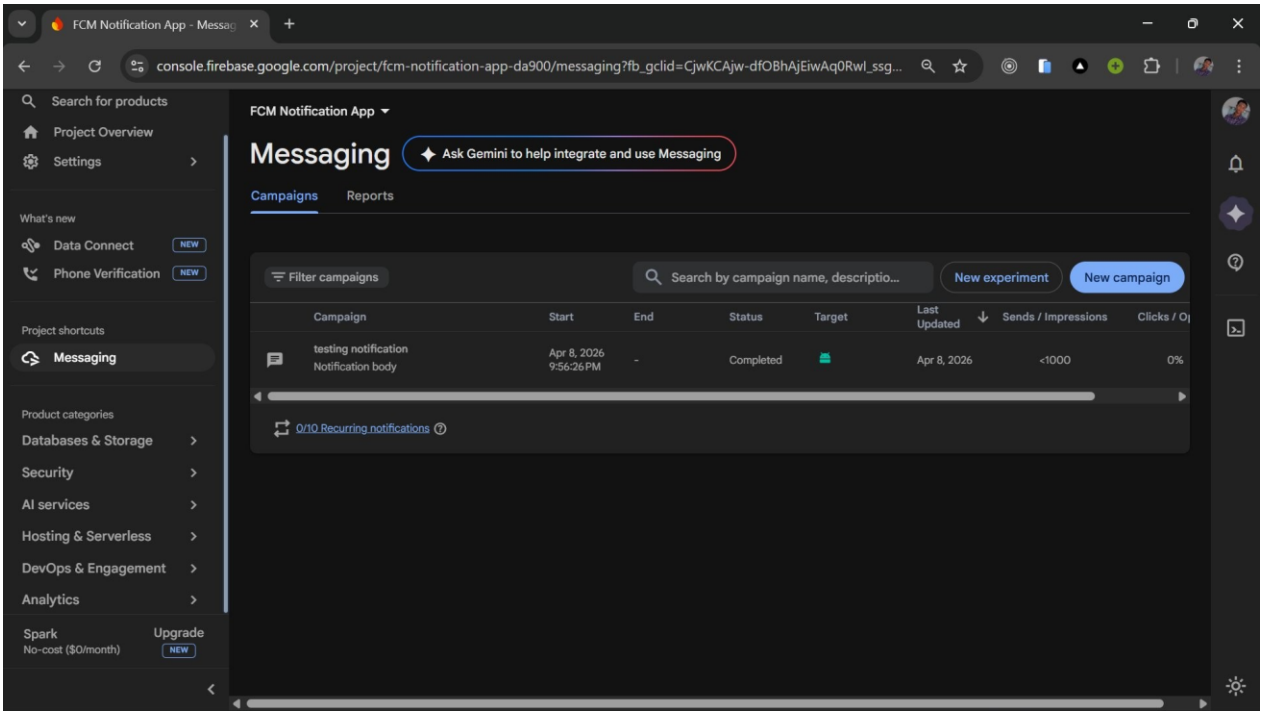

Figure 8: Firebase Console — Compose Notification screen with notification fields

4 Test on Device — Token Dialog

The Test on Device dialog is where the device FCM token is entered to send a targeted test notification. The dialog shows the recently used token from the Flutter app, which was selected and used to send the test.

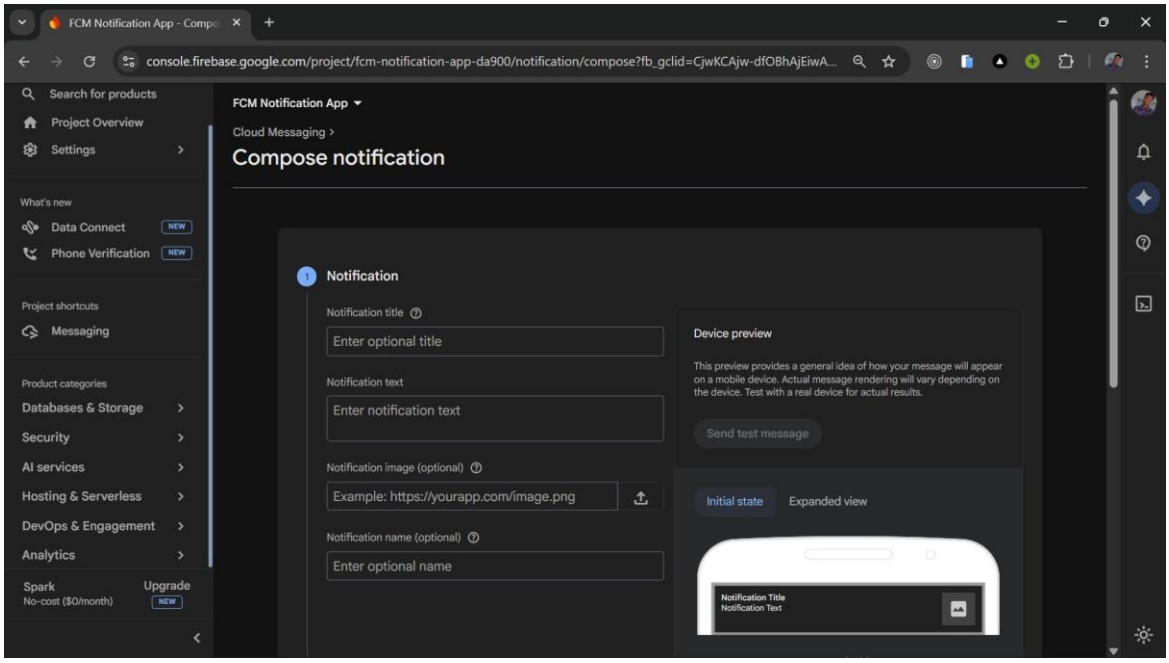

Figure 9: Test on Device dialog showing the FCM token from the Flutter app

5 Test on Device — FCM Token Dialog

The Test on Device dialog in Firebase Cloud Messaging Console, showing the FCM registration token retrieved from the Flutter application. The token (dpMvrfggR_WI3iJAg5-xbK:APA91bGgOIYRL...) appears under Recently Used and is selected to send a targeted test notification directly to the real Android device.

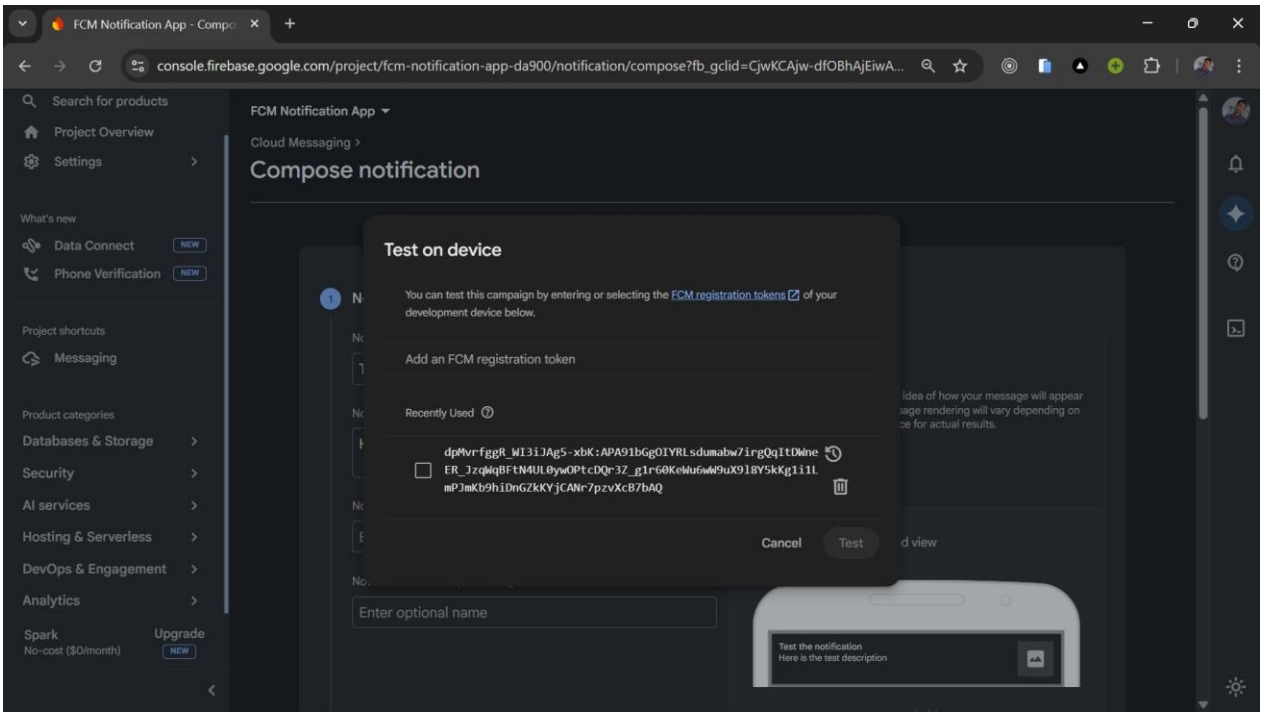

Figure 10: Firebase Project Settings showing the project configuration and registered Android app